# Confidence Intervals

Gov 2001: Quantitative Social Science Methods I

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# Welcome back — this week we stay in asymptotics

**Before break**: CLT, standard errors, consistency

**This week**:

- **Today**: Confidence intervals — the payoff of everything so far
- **Wednesday**: Practice problems together

**Next week**: Hypothesis testing

# Review: the CLT tells us the shape of $\bar{Y}_n$

From Week 7 — the result we'll use all day

If $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} (\mu, \sigma^2)$, then:

$$\frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

**In words**: no matter what the population looks like, the standardized sample mean is approximately normal for large $n$

**Why normal?** Each $Y_i$ carries its own idiosyncratic shape — skewness, bumps, gaps — but averaging washes those out, because they point in different directions across observations and cancel, leaving only symmetric variation around $\mu$

# Review: the standard error measures precision

From Week 7

**Standard error of $\bar{Y}_n$:**

$$\text{SE}(\bar{Y}_n) = \frac{\sigma}{\sqrt{n}}$$

**Problem**: we don't know $\sigma$

**Solution**: plug in $\hat{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2}$

This works because $\hat{\sigma} \xrightarrow{p} \sigma$ (consistency)

# Review: Slutsky says replacing $\sigma$ with $\hat{\sigma}$ is valid

From Week 7 — the step that makes the CLT usable

**We know**: $\dfrac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$

**We want**: to replace $\sigma$ (unknown) with $\hat{\sigma}$ (computed from data)

**Slutsky's theorem** says: if $\hat{\sigma} \xrightarrow{p} \sigma$, then

$$\boxed{\frac{\bar{Y}_n - \mu}{\hat{\sigma}/\sqrt{n}} \xrightarrow{d} N(0,1)}$$

Plugging in a consistent estimator doesn't change the limiting distribution

**This is the quantity we'll rearrange to build confidence intervals**

# Everything we've built leads to one place

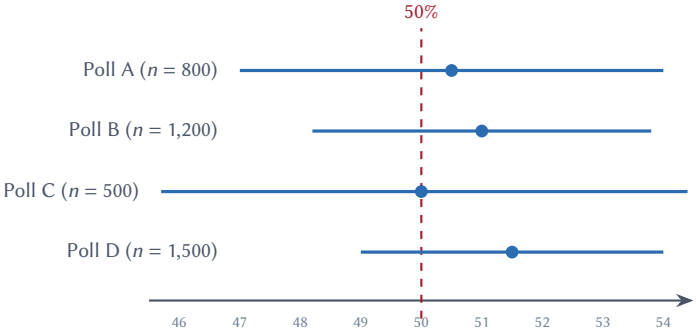
The logical chain from probability to inference

LLN: $\bar{Y}_n \xrightarrow{p} \mu$ → CLT: $\frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0,1)$ → **CI**: $\bar{Y} \pm z_{\alpha/2} \cdot \widehat{\text{SE}}$

*where* *what shape* *how precise*

# Polls always come with a margin of error

## November 2024 Pennsylvania polling



# Without uncertainty, point estimates are dangerous

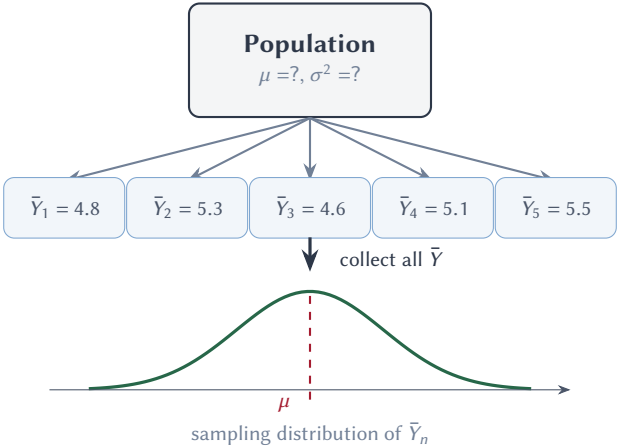
| Claim | Point estimate | 95% CI |
|---|---|---|
| Candidate leads by 2 pts | $\hat{p} = 0.52$ | $[0.489,\ 0.551]$ |
| GOTV raises turnout 5 pp | $\hat{\tau} = 0.05$ | $[-0.01,\ 0.11]$ |
| Job training raises wages \$2k | $\hat{\beta} = 2000$ | $[800,\ 3200]$ |

# Today's roadmap

1. **What is a confidence interval?**
   A range of values for the population parameter that are compatible with the data
2. **What is it not?**
   Not a probability statement about $\mu$ — it's a statement about the *procedure*
3. **How do we calculate one?**
   CLT + Slutsky $\longrightarrow$ $\bar{Y} \pm z_{\alpha/2} \times \widehat{\text{SE}}$
4. **What does it mean?**
   If we repeated the sampling, 95% of intervals built this way would contain $\mu$
5. **How should you use it in practice?**
   Report it alongside every point estimate — it tells your reader what the data can and cannot rule out

# Every sample gives a different answer

## The sampling distribution of $\bar{Y}_n$



# Variability is the problem — the CLT turns it into a solution

**Why not just report $\bar{Y}_n$?** Because it isn't $\mu$ — it's one realization from a distribution of possible sample means, and a different sample would give a different answer

**What we need**: to say how far $\bar{Y}_n$ could plausibly be from $\mu$

**What we need to know**: how much does $\bar{Y}_n$ vary, and in what shape?

**The CLT answers both**: the variation is $\sigma/\sqrt{n}$, and the shape is normal

That's enough to build an interval

# The CLT gives us a quantity we can rearrange

**From the CLT**:

$$Z_n = \frac{\bar{Y}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

*(this is the z-score — it measures how many SEs our sample mean is from the truth)*

**Slutsky** lets us plug in $\hat{\sigma} \xrightarrow{p} \sigma$:

$$\widehat{Z}_n = \frac{\bar{Y}_n - \mu}{\hat{\sigma}/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

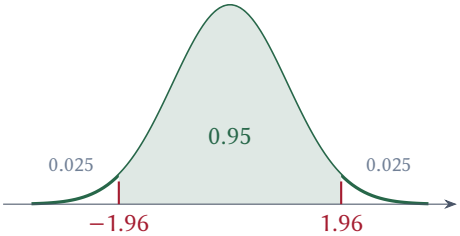
*(same thing, but now everything is computable except $\mu$)*

# This quantity has relatives — we'll meet them soon

| Quantity | Random? | What it does | When? |
|---|---|---|---|
| $\widehat{Z}_n = \dfrac{\bar{Y}_n - \mu}{\hat{\sigma}/\sqrt{n}}$ | Yes | Build CIs | **Today** |
| Test statistic: plug in $\mu_0$ for $\mu$ | Yes | Test claims about $\mu$ | Next week |
| $p$-value: tail area of test statistic | Yes | Quantify evidence | Next week |

Same machinery, different questions. Today we ask "where is $\mu$?" Next week we ask "is $\mu$ equal to some claimed value?"

# A standard normal concentrates 95% of its mass between $\pm 1.96$




*Since $\widehat{Z}_n \approx N(0,1)$ by the CLT, this 95% area applies to our quantity directly:*

$$\mathbb{P}\left(-1.96 < \frac{\bar{Y} - \mu}{\hat{\sigma}/\sqrt{n}} < 1.96\right) \approx 0.95$$

# Rearrange to isolate $\mu$

Three algebraic steps

**Start**: $\mathbb{P}\left(-1.96 < \frac{\bar{Y}-\mu}{\hat{\sigma}/\sqrt{n}} < 1.96\right) \approx 0.95$

**Multiply through** by $\hat{\sigma}/\sqrt{n}$:

$$\mathbb{P}\left(-1.96\,\frac{\hat{\sigma}}{\sqrt{n}} < \bar{Y} - \mu < 1.96\,\frac{\hat{\sigma}}{\sqrt{n}}\right) \approx 0.95$$

**Subtract** $\bar{Y}$, **multiply by** $-1$:

$$\mathbb{P}\left(\bar{Y} - 1.96\,\widehat{\mathrm{SE}} < \mu < \bar{Y} + 1.96\,\widehat{\mathrm{SE}}\right) \approx 0.95$$

# The confidence interval formula

**General formula** for level $1 - \alpha$:

$$\bar{Y} \pm z_{\alpha/2} \times \widehat{\text{SE}}$$

*$z_{\alpha/2}$ is the **critical value** — the cutoff that leaves $\alpha/2$ probability in each tail*

$$\boxed{95\%\ \text{CI}: \quad \bar{Y} \pm 1.96 \times \widehat{\text{SE}}}$$

$$99\%\ \text{CI}: \quad \bar{Y} \pm 2.576 \times \widehat{\text{SE}} \qquad 90\%\ \text{CI}: \quad \bar{Y} \pm 1.645 \times \widehat{\text{SE}}$$

Higher confidence → larger critical value → wider interval. The tradeoff is precision for coverage.

# Higher confidence requires a wider net

| Confidence level | $\alpha$ | $z_{\alpha/2}$ |
|---|---|---|
| 90% | 0.10 | 1.645 |
| 95% | 0.05 | 1.960 |
| 99% | 0.01 | 2.576 |

# Markov's inequality bounds tail probabilities using only the mean

Review from before break — you will use this on Problem Set 6

**Where we saw this**: Week 6, asymptotics — a tool for bounding probabilities without knowing the full distribution

**Why we need it now**: The CLT-based CI assumes normality (large $n$). Markov and Chebyshev give bounds that work *without* that assumption — a different route to uncertainty quantification

$$\text{If } Y \geq 0 \text{ with mean } \mathbb{E}[Y] = \mu: \quad \mathbb{P}(Y \geq a) \leq \frac{\mu}{a}$$

*You only need the mean. No variance, no normality, no large n.*

# Markov in practice: bounding hospital ER bills

**Setup**: ER bills are non-negative with $\mathbb{E}[Y] = \$800$

| Question | Bound | Calculation |
|---|---|---|
| $\mathbb{P}(Y \geq \$2{,}000)$ | $\leq 0.40$ | $800/2000$ |
| $\mathbb{P}(Y \geq \$4{,}000)$ | $\leq 0.20$ | $800/4000$ |
| $\mathbb{P}(Y \geq \$8{,}000)$ | $\leq 0.10$ | $800/8000$ |

These bounds are loose — but they're guaranteed. No assumptions about the shape of the distribution.

# Chebyshev adds the variance for a tighter bound

**Setup**: $Y$ with $\mathbb{E}[Y] = \mu$, $\text{Var}(Y) = \sigma^2$

$$\mathbb{P}(|Y - \mu| \geq k) \leq \frac{\sigma^2}{k^2}$$

*where $k$ is any distance from the mean — "how far from $\mu$ are we asking about?"*

**Same ER example**: $\mathbb{E}[Y] = \$800$, $\text{SD}(Y) = \$400$

$\mathbb{P}(|Y - 800| \geq 1{,}200) \leq \frac{400^2}{1200^2} = 0.111$

$\mathbb{P}(Y \geq 2{,}000) \leq 0.111$ vs. Markov: $\leq 0.40$

**Key**: more information (variance) $\longrightarrow$ tighter bound

# Chebyshev bounds tails even tighter — and that means we can build a CI

From bounding probabilities to bounding $\mu$

**Chebyshev bounds how far $\bar{Y}$ can fall from $\mu$:**

$$\mathbb{P}(|\bar{Y} - \mu| \geq k) \leq \frac{\text{Var}(\bar{Y})}{k^2}$$

**Why this matters for CIs**: this is already a statement about $|\bar{Y} - \mu|$ — the distance between our estimate and the truth. If we can bound that distance, we can build an interval around $\bar{Y}$ that traps $\mu$

## From Chebyshev to a confidence interval in three steps

1. **Choose** how much tail probability you'll tolerate ($\alpha$)
2. **Solve** for how far $\bar{Y}$ could plausibly be from $\mu$
3. **That distance is the margin of error** — no normality required

This is the same logic as the CLT-based CI: bound the distance between $\bar{Y}$ and $\mu$, then build an interval. The only difference is *what tool* gives us the bound.

# The Chebyshev CI: wider but assumption-free

**Set the bound equal to** $\alpha$ and solve for $k$: $\dfrac{\text{Var}(\bar{Y})}{k^2} = \alpha \implies k = \dfrac{\sigma}{\sqrt{n\alpha}}$

$$\text{Chebyshev } (1-\alpha) \text{ CI}: \quad \bar{Y} \pm \frac{\hat{\sigma}}{\sqrt{\alpha \cdot n}}$$

**At 95%**: multiplier $= 1/\sqrt{0.05} = 4.47$ vs. CLT's 1.96

**What this means**: you need ±4.47 standard errors to guarantee 95% coverage when you don't know the shape. Under the normal, ±1.96 is enough because the bell curve's tails die off fast

Less information about the distribution → wider interval. The normal assumption is powerful precisely because it rules out fat tails.

# The CLT gives a tighter CI by using more information

**CLT CI** (assumes large $n$, so $\bar{Y}$ is approximately normal):

$$\text{CLT } (1-\alpha) \text{ CI}: \quad \bar{Y} \pm z_{\alpha/2} \times \frac{\hat{\sigma}}{\sqrt{n}}$$

**At 95%**: multiplier = 1.96

| Method | Multiplier (95%) | Assumption |
|---|---|---|
| Chebyshev | 4.47 | Finite variance |
| CLT | 1.96 | Large $n$ |

## So why not always use the CLT?

**You should** — when $n$ is large enough for the normal approximation to be accurate

**Chebyshev is the guarantee that works even when it isn't**

- $n = 1{,}000$ poll? Use the CLT. The normal approximation is excellent
- $n = 15$ with a skewed distribution? Chebyshev is safer — the CLT may not have kicked in yet
- In practice, most applied work uses the CLT because samples are large

The real lesson: assumptions buy you precision. The CLT bets that $n$ is large enough — and that bet usually pays off ($4.47 \longrightarrow 1.96$).

## How large is "large $n$"?

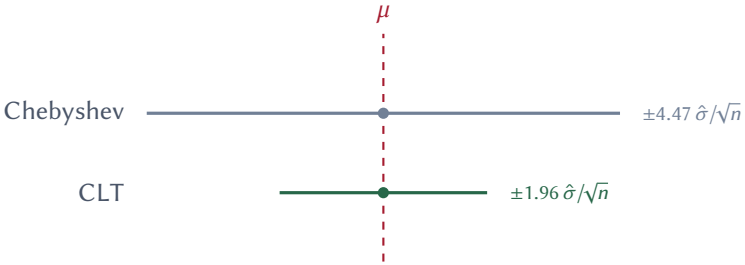
It depends on how non-normal the population is

| Population shape | $n$ needed | Example |
|---|---|---|
| Symmetric, light-tailed | 10–15 | Uniform |
| Moderately skewed | 30–50 | Exponential |
| Heavily skewed / heavy-tailed | 100+ | Log earnings, Pareto |
| Rare events ($p \approx 0.01$) | 500+ | Bernoulli(0.01) |

**The textbook rule**: $n \geq 30$ is a useful shorthand, not a law

The real driver is **skewness**. More skewed populations need larger $n$ for the CLT to kick in. The Berry-Esseen theorem makes this precise: approximation error is $O(1/\sqrt{n})$, with a constant that grows with the third moment.

# Chebyshev vs CLT: same center, very different width




**Tradeoff**: Chebyshev needs only finite variance; CLT needs large $n$

# The most common misinterpretation in statistics

**WRONG:** "There is a 95% probability that $\mu$ is in this interval."

**Correct**: 95% of intervals constructed this way contain $\mu$

| | |
|---|---|
| **What's fixed?** | $\mu$ — a number, not random |
| **What's random?** | The interval (depends on the sample) |
| **After computing:** | It either contains $\mu$ or it doesn't |

# This misinterpretation is so common it prompted a professional reckoning

**2016**: The American Statistical Association issues its first-ever statement on p-values — six principles clarifying what they do and don't mean

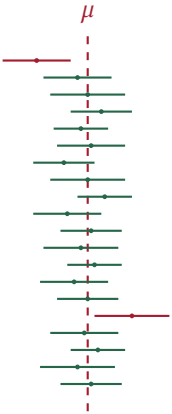
**2019**: Over 800 statisticians sign a call in *The American Statistician* to retire "statistical significance" as a binary threshold

**The problem isn't the tools** — it's that researchers routinely misinterpret them, and the misinterpretation has become the de facto interpretation in published work

This is why we drill the correct interpretation. Getting CIs and p-values right is an active problem in the profession — not a settled one.

# Twenty samples, twenty intervals — most catch the truth

Each horizontal line is one CI from a fresh i.i.d. sample




Green: contains $\mu$. Red: misses.

# The CI is a net, not a target

| | |
|---|---|
| **What's fixed?** | $\mu$ (a number, not random) |
| **What's random?** | The interval $(\bar{Y} \pm 1.96 \times \widehat{\text{SE}})$ |
| **Before sampling:** | 95% chance the net lands on $\mu$ |
| **After sampling:** | It either caught $\mu$ or it didn't |

## So which CI do you have?

You computed one interval from one sample. You can't know whether it's one of the 19 that caught $\mu$ or the 1 that missed

**That discomfort is real** — and it's the honest answer. A CI is not a guarantee about *this* interval. It's a guarantee about the *procedure*

**Why it's still useful**: you are betting on a method that works 95% of the time. You can't verify it in any single case — but repeatedly using a method with 95% coverage is a rational way to navigate uncertainty

It's like wearing a seatbelt. You can't know whether *this* drive is the one where it matters. But using the procedure is still the right call.

# Putting it into words: returns to a college degree

**Setup**: $\hat{\beta} = 15\%$, SE $= 7.5\%$, 95% CI: $[7.5\%,\ 22.5\%]$

**Say this:**

- "Our estimate of the return to college is 15%. If we repeated this study with a new i.i.d. sample, we'd expect different estimates — the SE of 7.5% tells us how much they'd typically vary"
- "The 95% CI of $[7.5\%, 22.5\%]$ says: values between 7.5% and 22.5% are *consistent with* our data. Values far outside — like 2% or 30% — are not. This is the range of plausible values for the true return, given this sample"
- "If we repeated this procedure many times, 95% of the resulting intervals would contain the true return — that's what justifies calling it a 95% CI"

# Putting it into words: what *not* to say

**Same setup**: $\hat{\beta} = 15\%$, SE $= 7.5\%$, 95% CI: $[7.5\%,\ 22.5\%]$

**Don't say this:**

- × "There is a 95% chance the true return is between 7.5% and 22.5%"
- × "The return to college is definitely 15%"
- × "The return is somewhere between 7.5% and 22.5%"

The first sounds right but treats $\mu$ as random. The second ignores uncertainty entirely. The third drops the "95%" and states it as a certainty. All three are published regularly.

# What are we estimating when we poll 1,000 voters?

Concept: identifying the population parameter

**The population**: all likely voters in this electorate

**The parameter**: $p$ = true proportion who support the candidate

**The sample**: $n = 1{,}000$ voters drawn i.i.d. from this population

**The estimate**: $\hat{p} = 0.52$ — this number depends on *which* 1,000 voters we happened to draw

**The question**: how far could $\hat{p}$ plausibly be from $p$?

# Example: polling margin of error

Calculate: SE $\rightarrow$ CI

**Setup**: Poll of $n = 1{,}000$ likely voters, $\hat{p} = 0.52$

**Step 1 — Standard error**:

$$\widehat{\text{SE}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.52 \times 0.48}{1000}} = 0.0158$$

**Step 2 — 95% CI**:

$$0.52 \pm 1.96 \times 0.0158 = [0.489,\ 0.551]$$

**Margin of error**: $1.96 \times 0.0158 \approx \pm 3.1$ percentage points

# Interpreting the polling CI

Interpret: what can we say?

$$[0.489,\ 0.551]$$

**The words**:

"If we were to repeatedly draw i.i.d. samples of 1,000 voters from this population and compute $\hat{p} \pm 1.96 \times \widehat{\text{SE}}$ each time, 95% of those intervals would contain the true population proportion $p$."

| | |
|---|---|
| **What's fixed?** | $p$ (the true proportion) |
| **What's random?** | The interval (it moves with each sample) |
| **This particular interval?** | Either contains $p$ or doesn't — we don't know |

# What are we estimating in a GOTV experiment?

## Concept: the population treatment effect

**The population**: all voters who could have received this mailer

**The parameter**: $\tau = p_T - p_C$ = the true difference in turnout rates between treatment and control populations

**The samples**: $n_T = 38{,}218$ treated, $n_C = 191{,}243$ control — drawn i.i.d. from their respective populations

**The estimate**: $\hat{\tau} = 0.081$ — a function of this particular sample

# Example: GOTV experiment

Gerber, Green, and Larimer (2008)

**Setup**: $n_T = 38{,}218$ (treatment), $n_C = 191{,}243$ (control)
$\hat{p}_T = 0.378, \quad \hat{p}_C = 0.297, \quad \hat{\tau} = 0.081$

**Step 1 — SE of the difference**:

$$\widehat{\text{SE}}(\hat{\tau}) = \sqrt{\frac{0.378 \times 0.622}{38{,}218} + \frac{0.297 \times 0.703}{191{,}243}} = 0.00264$$

**Step 2 — 95% CI**:

$$0.081 \pm 1.96 \times 0.00264 = [0.076,\ 0.086]$$

# Interpreting the GOTV CI

Interpret: what can we say?

$$[0.076,\ 0.086]$$

**The words**:

"If we were to repeatedly draw i.i.d. samples of this size from the treatment and control populations and compute $\hat{\tau} \pm 1.96 \times \widehat{\text{SE}}$ each time, 95% of those intervals would contain the true population difference in turnout $\tau$."

**Notice**: the interval is narrow ($\pm 0.5$ percentage points) because $n$ is enormous

**Notice**: the entire interval is far from zero — this effect is precisely estimated *and* substantively large

# Example: a smaller experiment tells a different story

What if we had less data?

**Similar effect size**, but $n_T = 500$, $n_C = 500$

$\hat{p}_T = 0.65, \quad \hat{p}_C = 0.60, \quad \hat{\tau} = 0.05$

**SE of the difference**:

$$\widehat{\mathsf{SE}}(\hat{\tau}) = \sqrt{\frac{0.65 \times 0.35}{500} + \frac{0.60 \times 0.40}{500}} = 0.0305$$

**95% CI**:

$$0.05 \pm 1.96 \times 0.0305 = [-0.010,\ 0.110]$$

## Same formula, very different conclusions

Interpret: sample size changes the story

| | Large experiment | Small experiment |
|---|---|---|
| $\hat{\tau}$ | 0.081 | 0.05 |
| $\widehat{\text{SE}}$ | 0.00264 | 0.0305 |
| 95% CI | $[0.076,\ 0.086]$ | $[-0.010,\ 0.110]$ |
| Excludes 0? | Yes | No |

**The lesson**: an effect can be real but undetectable with small $n$

The CI doesn't say "the effect is zero" — it says "we can't tell with this much data"

# What are we estimating in the canvassing experiment?

Concept: treatment effect on attitudes

**The population**: all people who could have been canvassed in this way

**The parameter**: $\tau = \mu_T - \mu_C$ = the true difference in mean attitude scores between treatment and control populations

**The samples**: $n_T \approx 1{,}100$ treated, $n_C \approx 1{,}050$ control — i.i.d. draws from their respective populations

**The estimate**: $\hat{\tau} = \bar{Y}_T - \bar{Y}_C$ — varies across samples

**The CI will tell us**: which values of $\tau$ are compatible with the data we observed

# CIs in the wild: canvassing and same-sex marriage attitudes

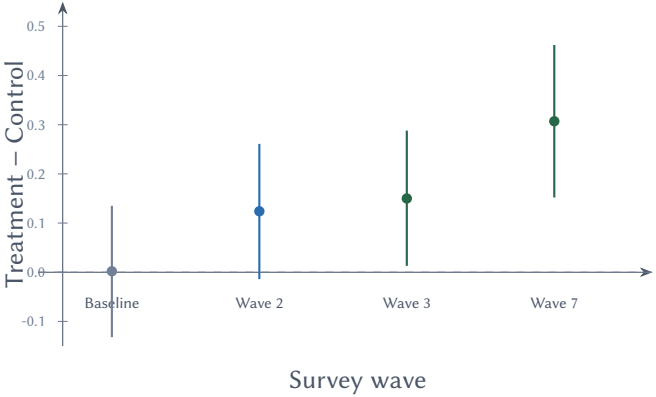
Broockman and Kalla (2016, *Science*)

**Background**: Door-to-door canvassing experiment

- Canvassers had conversations encouraging perspective-taking
- Treatment: same-sex marriage script ($n_T \approx 1{,}100$)
- Control: no contact ($n_C \approx 1{,}050$)
- Outcome: support for same-sex marriage (1–5 scale)

**Five survey waves**: baseline, then follow-ups over months

# The CIs tell a nuanced story — not every wave is significant




Gray: CI includes zero  Blue: borderline  Green: excludes zero

# Computing the CI: Wave 3 (3 days after canvassing)

Calculate: SE $\longrightarrow$ CI

**Wave 3**: $n_T = 1{,}058$, $n_C = 1{,}055$

$\bar{Y}_T = 3.102, \quad \bar{Y}_C = 2.952, \quad \hat{\tau} = 0.150$

**Step 1 — Standard error**:

$$\widehat{\mathsf{SE}}(\hat{\tau}) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}} = 0.070$$

**Step 2 — 95% CI**:

$$0.150 \pm 1.96 \times 0.070 = [0.013,\ 0.288]$$

# Computing the CI: Wave 7 (9 months after canvassing)

Calculate: SE $\longrightarrow$ CI

**Wave 7**: $n_T = 793$, $n_C = 735$

$\bar{Y}_T = 3.333, \quad \bar{Y}_C = 3.026, \quad \hat{\tau} = 0.307$

**Step 1 — Standard error**:

$$\widehat{\mathsf{SE}}(\hat{\tau}) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}} = 0.079$$

**Step 2 — 95% CI**:

$$0.307 \pm 1.96 \times 0.079 = [0.152,\ 0.462]$$

# Interpreting the Broockman and Kalla CIs

Interpret: what can we say?

**Wave 3** — CI: $[0.013,\ 0.288]$

"If we were to repeatedly draw i.i.d. samples of $\approx 1{,}050$ treated and control individuals from these populations and compute $\hat{\tau} \pm 1.96 \times \widehat{\mathsf{SE}}$ each time, 95% of those intervals would contain the true population difference in attitudes $\tau$."

**Wave 7** — CI: $[0.152,\ 0.462]$

Same interpretation. But the entire interval is far from zero — and the point estimate *grew* over time.

**The CIs tell us**: the effect is real, it persists, and it may even strengthen

# A note on where the randomness lives

An aside for the intellectually curious

We just said: "if we repeatedly sampled…"

But these 2,113 people aren't drawn from a super-population — they are the actual participants. The randomness came from the coin flip that assigned treatment.

| | **Our framework** | **Design-based** |
|---|---|---|
| Randomness | Sampling from $P$ | Treatment assignment |
| Outcomes | Random draws | Fixed numbers |
| Target | Population $\mu_T - \mu_C$ | Finite-pop. EATT |
| CI formula | *the same* | |

Rambachan and Roth (2025, *JASA*) formalize this. The formula you just learned is more robust than the derivation suggests. For this course: we stay with sampling.

# Three levers control the width of a confidence interval

$$\text{Width} = 2 \times z_{\alpha/2} \times \frac{\hat{\sigma}}{\sqrt{n}}$$

**Confidence level**

$z_{\alpha/2}$

Higher $\longrightarrow$ wider

**Variability**

$\hat{\sigma}$

Noisier $\longrightarrow$ wider

**Sample size**

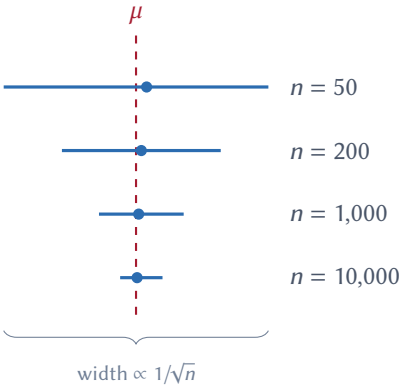
$n$ (via $1/\sqrt{n}$)

More data

$\longrightarrow$ narrower

# Visualizing the sample size effect

Same population, same $\mu$, different $n$



# How much data do you need?

## Sample size planning

**Goal**: 95% CI with margin of error $\leq m$

**Requirement**: $1.96 \times \sigma / \sqrt{n} \leq m$

$$n \geq \left( \frac{1.96\,\sigma}{m} \right)^2$$

**Polling example**: $\pm 3\%$ margin of error, worst-case $\sigma = 0.5$:

$$n \geq \left( \frac{1.96 \times 0.5}{0.03} \right)^2 = 1{,}067$$

# The $\sqrt{n}$ rule makes large effects cheap and small effects expensive

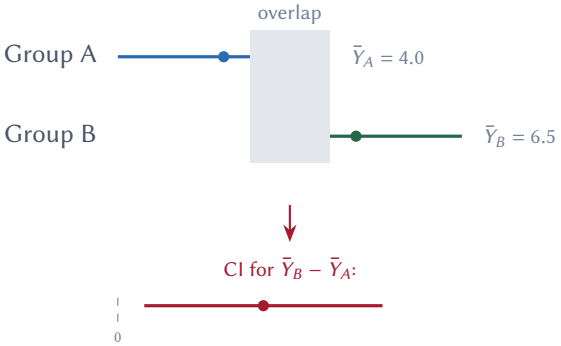
| Desired MoE | Required $n$ | Context |
|---:|---:|---|
| ±10% | 96 | Pilot study |
| ±5% | 384 | Departmental survey |
| ±3% | 1,067 | National poll |
| ±1% | 9,604 | Large-scale experiment |
| ±0.5% | 38,416 | Census-scale |

# Three common mistakes about confidence intervals

| Mistake | Correct |
|---|---|
| "95% probability $\mu \in [a, b]$" | 95% of intervals contain $\mu$ |
| "Overlapping CIs $\Rightarrow$ no difference" | CIs can overlap even when the difference is significant |
| "Wider CI = worse estimate" | Width reflects $n$ and $\sigma$, not estimator quality |

# Overlapping CIs do not mean "no difference"

A surprisingly common error in published research



# What confidence intervals can and cannot tell you

**CIs can**:

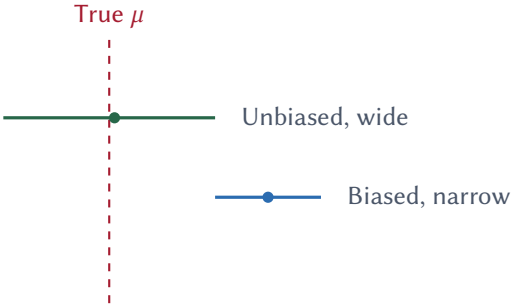
- Quantify sampling uncertainty around an estimate
- Show which values of $\mu$ are compatible with the data
- Guide sample size decisions before data collection
- Compare groups via the CI for their *difference*

**CIs cannot**:

- Fix a biased estimator (garbage in, precise garbage out)
- Account for non-sampling errors (measurement, selection)
- Tell you whether an effect is *practically* important
- Replace thinking about *identification*

# A narrow CI around a biased estimate is worse than useless

Precision ≠ accuracy



# The delta method extends CIs to transformations of $\mu$

Review from before break

**Problem**: We want a CI for $g(\mu)$, not $\mu$ itself

**Delta method**: if $\sqrt{n}(\bar{Y} - \mu) \xrightarrow{d} N(0, \sigma^2)$ and $g'(\mu) \neq 0$:

$$\boxed{\sqrt{n}\big(g(\bar{Y}) - g(\mu)\big) \xrightarrow{d} N\big(0,\ [g'(\mu)]^2\sigma^2\big)}$$

**CI for** $g(\mu)$:

$$g(\bar{Y}) \ \pm \ z_{\alpha/2} \times |g'(\bar{Y})| \times \widehat{\text{SE}}(\bar{Y})$$

# Delta method example: CI for $\mu^2$

Anticipating Problem Set 6

**Setup**: $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\mu, \sigma^2)$, want CI for $\theta = \mu^2$

$g(\mu) = \mu^2, \quad g'(\mu) = 2\mu$

**By CLT**: $\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$

**By delta method**:

$$\sqrt{n}(\bar{X}^2 - \mu^2) \xrightarrow{d} N(0,\ 4\mu^2\sigma^2)$$

**95% CI for $\mu^2$**:

$$\bar{X}^2 \ \pm\ 1.96 \times \frac{2|\bar{X}|\,\hat{\sigma}}{\sqrt{n}}$$

# Delta method example: CI for log odds

From Bernoulli data to logistic regression

**Setup**: $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bern}(p)$, want CI for $\theta = \log\left(\frac{p}{1-p}\right)$

$g(p) = \log\left(\frac{p}{1-p}\right), \quad g'(p) = \frac{1}{p(1-p)}$

**By CLT**: $\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0,\ p(1-p))$

**By delta method**:

$$\sqrt{n}\big(g(\hat{p}) - g(p)\big) \xrightarrow{d} N\left(0,\ \frac{1}{p(1-p)}\right)$$

**95% CI for log odds**: $g(\hat{p}) \ \pm \ \frac{1.96}{\sqrt{n \cdot \hat{p}(1-\hat{p})}}$

# CIs from the MLE use Fisher information

Connecting likelihood (Week 6) to inference (today)

**Recall**: the MLE $\hat{\theta}_{\text{MLE}}$ is asymptotically normal:

$$\sqrt{n}\left(\hat{\theta}_{\text{MLE}} - \theta_0\right) \xrightarrow{d} N\left(0,\ \frac{1}{I(\theta_0)}\right)$$

**Fisher information**: $I(\theta) = -\mathbb{E}\left[\frac{d^2}{d\theta^2} \log f(X;\theta)\right]$

**MLE confidence interval**:

$$\hat{\theta}_{\text{MLE}} \ \pm\ z_{\alpha/2} \times \frac{1}{\sqrt{n \cdot I(\hat{\theta}_{\text{MLE}})}}$$

# Example: CI for the Poisson rate

Anticipating Problem Set 6

**Setup**: $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Pois}(\lambda)$

**Log-likelihood**: $\ell(\lambda) = (\sum x_i) \log \lambda - n\lambda + C$

**MLE**: $\hat{\lambda} = \bar{X}$ (take FOC, solve)

**Fisher information**: $I(\lambda) = 1/\lambda$

**95% CI**:

$$\bar{X} \pm 1.96 \times \frac{1}{\sqrt{n/\hat{\lambda}}} = \bar{X} \pm 1.96 \times \sqrt{\frac{\hat{\lambda}}{n}} = \bar{X} \pm 1.96 \times \sqrt{\frac{\bar{X}}{n}}$$

# Example: Bernoulli CI both ways

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bern}(p), \quad \hat{p} = \bar{X}$

**Setup**: $n = 400$, $\hat{p} = 0.55$, $\hat{\sigma}^2 = \hat{p}(1 - \hat{p}) = 0.2475$

**Chebyshev 95% CI**:

$$0.55 \pm \frac{\sqrt{0.2475}}{\sqrt{0.05 \times 400}} = 0.55 \pm \frac{0.4975}{4.472} = 0.55 \pm 0.111 = [0.439,\ 0.661]$$

**CLT 95% CI**:

$$0.55 \pm 1.96 \times \frac{\sqrt{0.2475}}{\sqrt{400}} = 0.55 \pm 1.96 \times 0.0249 = 0.55 \pm 0.049 = [0.501,\ 0.599]$$

# Wednesday: practice problems together

**We will work through problems like the ones on Problem Set 6**:

- Markov and Chebyshev bounds (Pset 6, Q2)
- Consistency proofs via LLN (Pset 6, Q3)
- CLT and delta method for asymptotic distributions (Pset 6, Q3–5)
- Building CIs from scratch: Chebyshev vs CLT (Pset 6, Q6)

**Bring**: pen, paper, calculator (or laptop with R)

**Pset 6 due**: Tuesday Mar 31 | **Next week**: Hypothesis testing

# The full toolkit: five results, one inference machine

**LLN**: $\bar{Y}_n \xrightarrow{p} \mu$

**CMT**: $g(\bar{Y}_n) \xrightarrow{p} g(\mu)$

**CLT**: $\sqrt{n}(\bar{Y}_n - \mu) \xrightarrow{d} N(0, \sigma^2)$

**Slutsky**: plug in $\hat{\sigma} \xrightarrow{p} \sigma$

**Delta**: $\sqrt{n}(g(\bar{Y}) - g(\mu)) \xrightarrow{d} N(0, [g']^2\sigma^2)$

**Confidence Intervals**